

ECEN-601: Homework 1

Due on September 12, 2025 at 5:00 pm

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Problem 1

Which of the following expressions are statements?

- (a) Today is a nice day. (**Not a statement**)
- (b) Go to sleep. (**Not a statement**)
- (c) Is it going to snow tomorrow? (**Not a statement**)
- (d) The U.S. has 49 states. (**Statement**)
- (e) I like to eat fruit, and you often think about traveling to Spain. (**Not a statement**)
- (f) If we go out tonight, the babysitter will be unhappy. (**Not a statement**)
- (g) Call me on Thursday if you are home. (**Not a statement**)

Solution

Only **(d)** is a statement.

I feel like **(a)** *could* be a statement, though it varies from person-to-person (similar to **(e)**), so I say not a statement. Additionally, **(f)** is a conditional statement, but I don't think that's what the question is asking.

Problem 2

Let X = "I am happy," let Y = "I am watching a movie" and Z = "I am eating spaghetti." Translate the following statements into words.

- (a) $Z \rightarrow X$.
- (b) $X \leftrightarrow Y$.
- (c) $(Y \vee Z) \rightarrow X$.
- (d) $Y \vee (Z \rightarrow X)$.
- (e) $(Y \rightarrow \neg X) \wedge (Z \rightarrow \neg X)$.
- (f) $(X \wedge \neg Y) \leftrightarrow (Y \vee Z)$.

Solution

Part (a): If I am eating spaghetti, then I am happy.

Part (b): I am happy if and only if I am watching a movie.

Part (c): If I am watching a movie or I am eating spaghetti, then I am happy.

Part (d): If I am eating spaghetti then I am happy, or I am watching a movie.

Part (e): If I am watching a movie then I am not happy, and if I am eating spaghetti then I am not happy.

Part (f): I am happy and I am not watching a movie, if and only if I am watching a movie or I am eating spaghetti.

Problem 3

Let $E =$ "The house is blue," let $F =$ "The house is 30 years old" and $G =$ "The house is ugly."
Translate the following statements into symbols.

- (a) If the house is 30 years old, then it is ugly.
- (b) If the house is blue, then it is ugly or it is 30 years old.
- (c) If the house is blue, then it is ugly, or it is 30 years old.
- (d) The house is not ugly if and only if it is 30 years old.
- (e) The house is 30 years old if it is blue, and it is not ugly if it is 30 years old.
- (f) For the house to be ugly, it is necessary and sufficient that it be ugly and 30 years old.

Solution

Part (a): $F \rightarrow G$.

Part (b): $E \rightarrow (G \vee F)$

Part (c): $(E \rightarrow G) \vee F$

Part (d): $\neg G \leftrightarrow F$

Part (e): $(E \rightarrow F) \wedge (F \rightarrow \neg G)$

Part (f): $G \leftrightarrow (G \wedge F)$

Problem 4

Make a truth table for each of the following statements.

- (a) $P \wedge \neg Q$.
- (b) $(R \vee S) \wedge \neg R$.
- (c) $X \vee (\neg Y \vee Z)$.
- (d) $(A \vee B) \wedge (A \vee C)$.
- (e) $(P \wedge R) \vee \neg(Q \wedge S)$.

Solution

I have constructed these truth tables in a manner similar to the truth tables in Engineering Fundamentals. Additionally, I have highlighted the final column of each truth table in light green.

Part (a)

P	Q	P	$\wedge$	$\neg Q$
T	T	T	T	F
T	F	T	F	T
F	T	F	F	F
F	F	F	F	T

Part (b)

R	S	$(R \vee S)$	$\wedge$	$\neg R$
T	T	T	F	F
T	F	T	F	F
F	T	F	T	T
F	F	F	F	T

Part (c)

X	Y	Z	X	$\vee$	$(\neg Y \vee Z)$
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	F	T	T
F	T	F	F	F	F
F	F	T	F	T	T
F	F	F	F	T	F

Part (d)

A	B	C	$(A \vee B)$	$\wedge$	$(A \vee C)$
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	F	T	T
F	T	F	F	F	F
F	F	T	F	F	T
F	F	F	F	F	F

Final truth table on following page...

Part (e)

P	Q	R	S	$(P \wedge R)$	$\vee$	$\neg$	$(Q \wedge S)$
T	T	T	T	T	T	F	T
T	T	T	F	T	T	T	F
T	T	F	T	F	F	T	T
T	T	F	F	F	T	T	F
T	F	T	T	T	T	F	F
T	F	T	F	T	T	F	F
T	F	F	T	F	T	F	F
T	F	F	F	F	T	T	F
F	T	T	T	F	F	T	T
F	T	T	F	F	T	T	F
F	T	F	T	F	F	T	T
F	T	F	F	F	T	T	F
F	F	T	T	F	T	F	F
F	F	T	F	F	T	F	F
F	F	F	T	F	T	F	F
F	F	F	F	F	T	T	F

Problem 5

Simplify the following statements (making use of any equivalence of statements given so far in the text or exercises).

- (a) $\neg(P \rightarrow \neg Q)$
- (b) $A \rightarrow (A \wedge B)$
- (c) $(X \wedge Y) \rightarrow X$
- (d) $\neg(M \vee L) \wedge L$
- (e) $(P \rightarrow Q) \vee Q$
- (f) $\neg(X \rightarrow Y) \vee Y$

Solution

These simplifications can be found pretty quickly by writing out the truth tables, though I will omit that process and use the stated identities in the text and notes.

Part (a): $\neg(P \rightarrow \neg Q) \equiv P \wedge \neg(\neg Q) \equiv P \wedge Q.$

Part (b): $A \rightarrow (A \wedge B) \equiv \neg A \vee (A \wedge B) \equiv (\neg A \vee A) \wedge (\neg A \vee B) \equiv \neg A \vee B.$

Part (c): $(X \wedge Y) \rightarrow X \equiv \neg(X \wedge Y) \vee X \equiv \underbrace{(\neg X \vee X)}_{\text{tautology}} \vee (\neg Y) \equiv \text{TRUE}.$

Part (d): $\neg(M \vee L) \wedge L \equiv (\neg M \wedge \neg L) \wedge L \equiv \underbrace{(\neg L \wedge L)}_{\text{contradiction}} \wedge L \equiv \text{FALSE}.$

Part (e): $(P \rightarrow Q) \vee Q \equiv (\neg P \vee Q) \vee Q \equiv (Q \vee Q) \vee \neg P \equiv Q \vee \neg P$

Part (f): $\neg(X \rightarrow Y) \vee Y \equiv (X \wedge \neg Y) \vee Y \equiv \underbrace{(Y \vee \neg Y)}_{\text{tautology}} \wedge (Y \vee X) \equiv X \vee Y$

Problem 6

Suppose that the possible values of x are all people. Let $Y(x)$ = “ x has green hair,” let $Z(x)$ = “ x likes pickles” and $W(x)$ = “ x has a pet frog.” Translate the following statements into words.

- (a) $(\forall x)Y(x)$.
- (b) $(\exists x)Z(x)$.
- (c) $(\forall x)[W(x) \wedge Z(x)]$.
- (d) $(\exists x)[Y(x) \rightarrow W(x)]$.
- (e) $(\forall x)[W(x) \leftrightarrow \neg Z(x)]$.

Solution

Part (a): Every person has green hair.

Part (b): At least one person likes pickles. // Some people like pickles.

Part (c): All people have a pet frog and like pickles.

Part (d): For some people, if they have green hair, then they have a pet frog.

Part (e): Every person has a pet frog if and only if they do not like pickles.